AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

Multi-Hazard Mapping using VLM

A Vision-Language Framework for Multi-Hazard Mapping on Earth Observation Data

K. Siva Pranav – CB.SC.U4AIE23309

K. Mohana Venkata Krishna – CB.SC.U4AIE23313

G. Thanuj – CB.SC.U4AIE23342

P. Tejadeep – CB.SC.U4AIE23328

Team Number: D15

Supervisor: A. Shashank

Co-supervisor: Dr. T. Keerthika

1. Introduction

- Floods, Wildfire, and landslides cause escalating loss of life and infrastructure, and traditional hazard specific field monitoring and single day forecasting are too slow, costly, and fragmented to support continuous, forward looking risk tracking across hazard types. [9]

- Recent VLM and deep learning work has advanced disaster perception through classification and segmentation backbones (Mamba, ViT, ConvNeXT) and shown that fusing multimodal satellite and radar imagery with ancillary geospatial context consistently outperforms single modality, RGB only approaches. [2][3][4]

- Despite this progress, existing systems stop at post event classification or static change detection, leaving a gap between raw model outputs and continuous, decision ready hazard intelligence; current VLMs also lack the spatial regression needed to project a hazard's evolving impact radius over time.

CASE STUDY

Uttarakhand 2024 Forest Fire Season, Kumaon Division

Baseline conditions: A prolonged dry spell through March 2024, following weak winter rainfall, depleted fuel and soil moisture across Nainital, Almora, and Champawat districts.

Cascade observed: Forest fire frequency in Uttarakhand between 1 February and 24 April 2024 more than doubled compared to the same period in 2023, with Nainital district alone recording roughly five times more incidents year-on-year. Fire activity peaked in two waves (early and late April), tracked in near real time via NASA FIRMS (MODIS/VIIRS) and burn-severity mapping from Sentinel-2. [5][6]

Gap illustrated: FSI and NASA FIRMS deliver reliable near real-time hotspot detection, but neither system forecasts how a fire's burn scar and downstream landslide susceptibility (via loss of root-reinforcement on steep Himalayan slopes) will evolve over the following weeks the exact continuous, multi-day impact-radius forecasting gap MultiHazard-VLM is designed to fill. [21]

2. Literature Review

Paper	Author	Method	Dataset	Results	Limitation
[1] **DisasterM3** 2025	J. Wang et al.	Fuses optical + SAR imagery across 9 disaster perception tasks with general-purpose VLMs	Multi-hazard optical + SAR remote-sensing benchmark	Strong performance on single-hazard perception tasks	Fails zero-shot generalization on multi-hazard cascades; hallucinates under cloud cover
[2] **FloodSense-VLM** 2026	D. S. P. Govindaraj et al.	Lightweight edge-deployable VLM for visual QA and real-time flood segmentation	Localized flood imagery + VQA annotation set	Real-time inference feasible on edge hardware	Short context window; cannot ingest long-term historical time-series data
[3] **SeaMo** 2025	X. Li et al.	Season-aware fusion of spatial, multimodal and multi-seasonal satellite arrays	Multi-seasonal satellite image archive	Robust cross-season feature representations	Passive feature extractor; no predictive forecasting module
[4] **Modality-Resilient EO Foundation Model** 2026	A. K. Pollan	Cross-attention masking to handle missing/corrupted sensor streams	Multi-sensor Earth-observation data with induced sensor dropout	Maintains accuracy under missing optical/SAR channels	Limited to static change detection, not real-time forecasting

2. Literature Review

Paper	Author	Method	Dataset	Results	Limitation
[5] **Mamba-MDRNet** 2025	Y. Li et al.	LLM-extracted text descriptors fused with visual features via selective Mamba scan	Labeled multimodal disaster scene dataset	97.8% scene classification accuracy	Classifies scene type only; no spatial or damage-radius regression
[6] **Multi-Modal Attention for Damage Classification** 2026	R. Zhang et al.	Bi-temporal ConvNeXT-Tiny features with cross-attention for damage-level isolation	Pre- and post-disaster paired satellite imagery	High infrastructure-damage classification accuracy	Needs post-disaster imagery; unusable as an early-warning tool
[7] **Foundation Model for Heterogeneous Disaster Detection** 2026	S. L. Jenkins et al.	Maps headlines, social streams and weather alerts onto an H3 spatial grid via generative reasoning	Text headlines, social-media streams, weather-alert feeds	Effective fusion of heterogeneous unstructured sources	No pixel-level CV binding; cannot measure physical footprints
[8] **Explainable Disaster Detection (ViT + Grad-CAM++/LIME)** 2024	A. Sharma et al.	Vision Transformer combined with Grad-CAM++ and LIME for explainable classification	Labeled natural-disaster image corpus	Interpretable, human-readable classification outputs	Static, non-interoperable labels; cannot trace sequential/overlapping hazards

3. Research Gap

Three gaps distilled from the reviewed literature

1

The Computational Lag Gap (Architecture Level): Current Vision-Language Models use standard Transformer blocks. These networks suffer from massive computational bottlenecks, making them too slow to process a continuous 30-day rolling timeline of environmental changes. [7]

2

The Spatial Blindness Gap (Spatial Level): Existing systems are designed for broad, macro-level alerts (e.g., "heavy rain across North-East India"). They lack dedicated spatial regression layers capable of pinpointing a localized threat epicenter or calculating an exact physical destruction radius (Rh) in kilometers. [7]

3

The Logical Inference Gap (Output Level): Generative VLMs try to output text descriptions and geographic coordinates together through a single text engine. This unconstrained text generation causes the model to "hallucinate" numbers, creating fabricated coordinates that are dangerous for emergency deployments. [7]

4. Problem Statement

- Traditional field-based landslide assessments are costly, slow, and unsafe in hazardous terrain, while existing deep learning models only focus on isolated detection or mapping.
- A critical gap remains between raw AI outputs and the actionable, structured intelligence required for rapid disaster management.
- Current remote sensing Vision-Language Models (VLMs) produce generic image captions but lack terrain-based geospatial reasoning.
- Standard image-level interpretations frequently suffer from hallucinations because they ignore key controlling factors like topography, slope, and lithology.
- Manual post-event workflows still heavily rely on experts to manually interpret fragmented data and compile operational reports under time-sensitive conditions. [8]

Scope Boundaries

- **Hazards covered:** Floods, wildfire, and rainfall-induced landslides
- **Geographic scope:** India-focused deployment, generalizable to any region with equivalent data feeds
- **Spatial output:** Hazard epicenter coordinates + dynamic impact radius (0.1 to 30 km, clamped)
- **Out of scope:** Seismic earthquake prediction, post-disaster relief logistics

5. Objectives

1. **Develop an Automated End-to-End Pipeline:** Bridge the gap between satellite data acquisition and report generation to deliver actionable geohazard intelligence for post-disaster response.

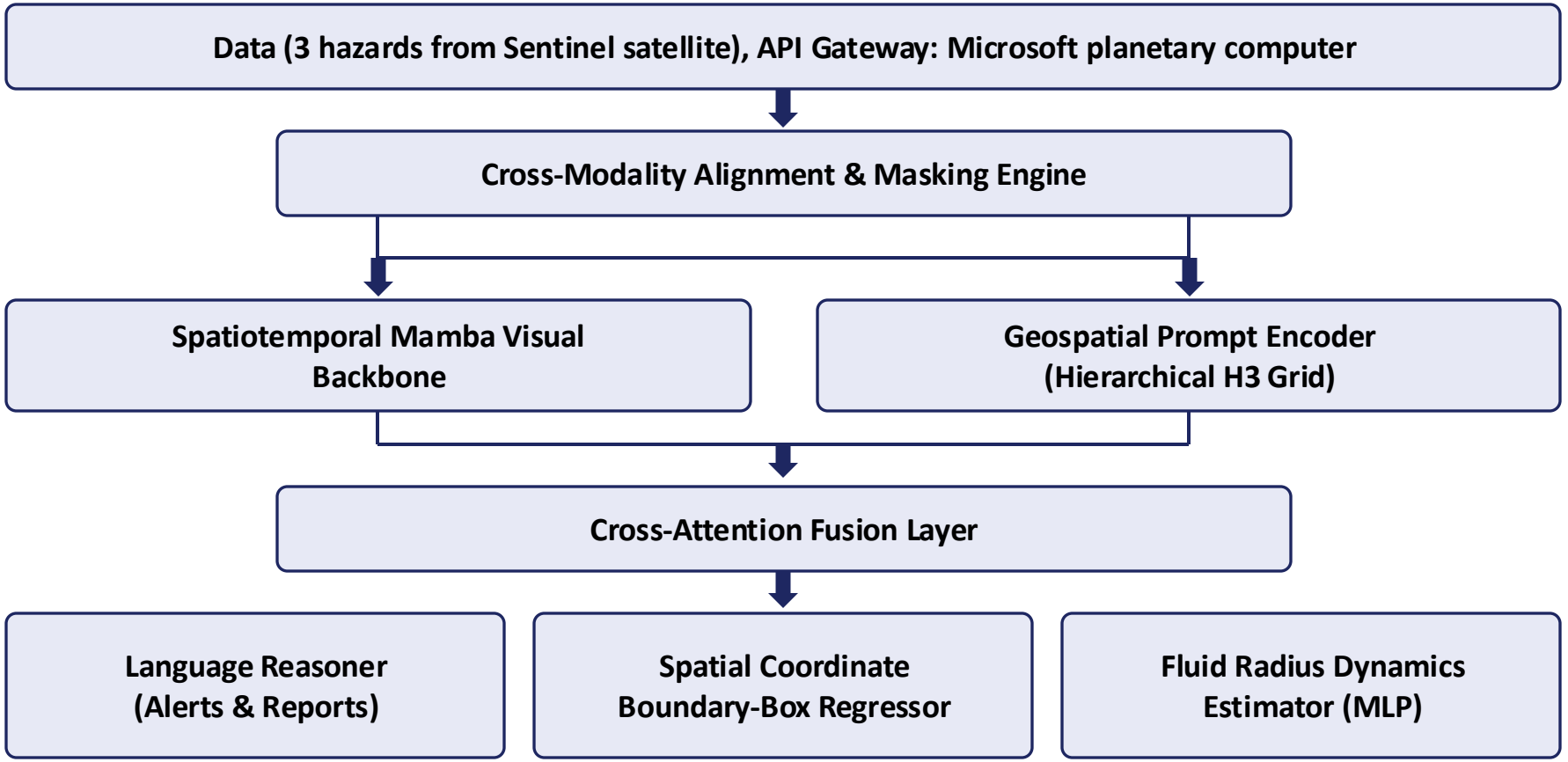
2. **Enhance Pixel-Level Localization:** Improve landslide footprint segmentation accuracy by engineering an optimized SDF-SegFormer-B2 architecture integrated with a combined Focal–Dice loss and an auxiliary slope channel.

3. **Extract Geospatial & Physical Context:** Ingest ancillary geospatial layers (DEM, slope, NDVI, lithology) using threshold-based techniques to enrich detected regions with key terrain attributes.

4. **Implement a Geospatially Grounded LLM:** Condition a Large Language Model directly on extracted geo-attributes to eliminate vision-language hallucinations and ensure physical consistency.

5. **Generate Actionable Expert Reports:** Automatically produce structured, expert-style reports featuring geospatial characterizations, analytical interpretations, and prioritized intervention recommendations

6. Proposed Methodology / Framework



7. Datasets & Data Feeds

Global S1S2-Water Benchmark Dataset

- **Sentinel-1 (S1):** Includes Ground Range Detected (GRD) products with 2 bands (VV and VH) in dB units (scaled by 100).
- **Sentinel-2 (S2):** Includes Level-1C (L1C) products with 6 bands (Blue, Green, Red, NIR, SWIR1, and SWIR2) measuring Top-Of-Atmosphere (TOA) reflectance (scaled by 10,000).
- Both image products are stored as Cloud Optimized GeoTIFFs (COG) with a resolution of 10980 × 10980 pixels projected to UTM.[14][15]

Dataset Card for HLS Burn Scar Scenes

- Purpose & Contents: Features Harmonized Landsat and Sentinel-2 (HLS) 6-band satellite imagery paired with single-band burn masks (2018–2021) over the contiguous United States, designed specifically for training geospatial machine learning models.
- Format & Scale: Contains 804 scenes (512x512 pixels) split into training and validation sets, covering burn scars (11%), unburned areas (88%), and missing data (1%). [14]

CAS Landslide Dataset

- A Large-Scale and multi-sensor collection of 20,865 high-resolution (512x512 pixels) RGB images combining satellite and unmanned aerial vehicle (UAV) imagery across nine distinct regions. Developed by the Chinese Academy of Sciences, it provides quality-filtered, standardized benchmark data with precise pixel-level masks to support deep learning-based landslide detection and hazard monitoring. [14]

Microsoft Planetary Computer

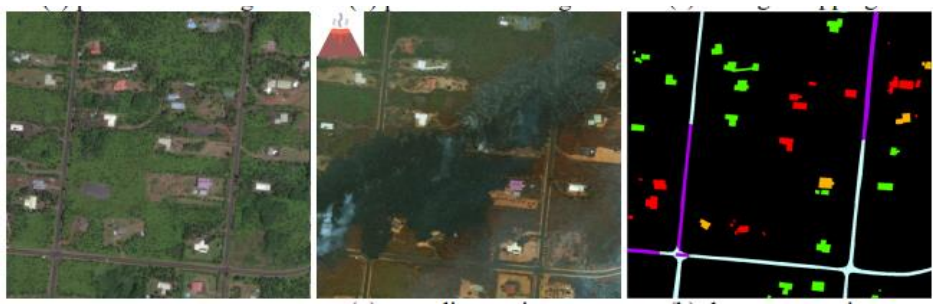
- Geospatial Analytics Platform: It provides APIs, computing resources, and developer tools (built on Microsoft Azure) that allow researchers, conservationists, and organizations to query, analyze, and build applications for climate action and sustainability decision-making. [14]

7. Dataset & Data Feeds

Dataset Construction Pipeline



Dataset images





- **Visualizing Temporal Impact:** The sample images pair pre-disaster baseline satellite/aerial imagery with post-disaster captures to highlight structural and environmental changes over time.

- **Geospatial & Damage Mapping:** They incorporate classified segmentation masks and NDVI/SAR overlay layers to precisely pinpoint localized damage severity and land-surface alterations for VLM training. [9]

8. Techniques & Tools

Algorithms & Model Components

- Spatiotemporal Mamba selective-scan visual backbone
- Cross-attention fusion of visual and text-prompt tokens
- Hierarchical H3 geospatial grid encoding
- Softmax hazard-probability classification head
- Sigmoid coordinate regression head + ReLU radius regression head

Software & Deployment Platforms

- Python / PyTorch model development
- Docker + Kubernetes microservice orchestration
- REST API gateway with automated cron-based ingestion
- Uber H3 library for spatial indexing and heatmap rendering

9. Expected Outcomes

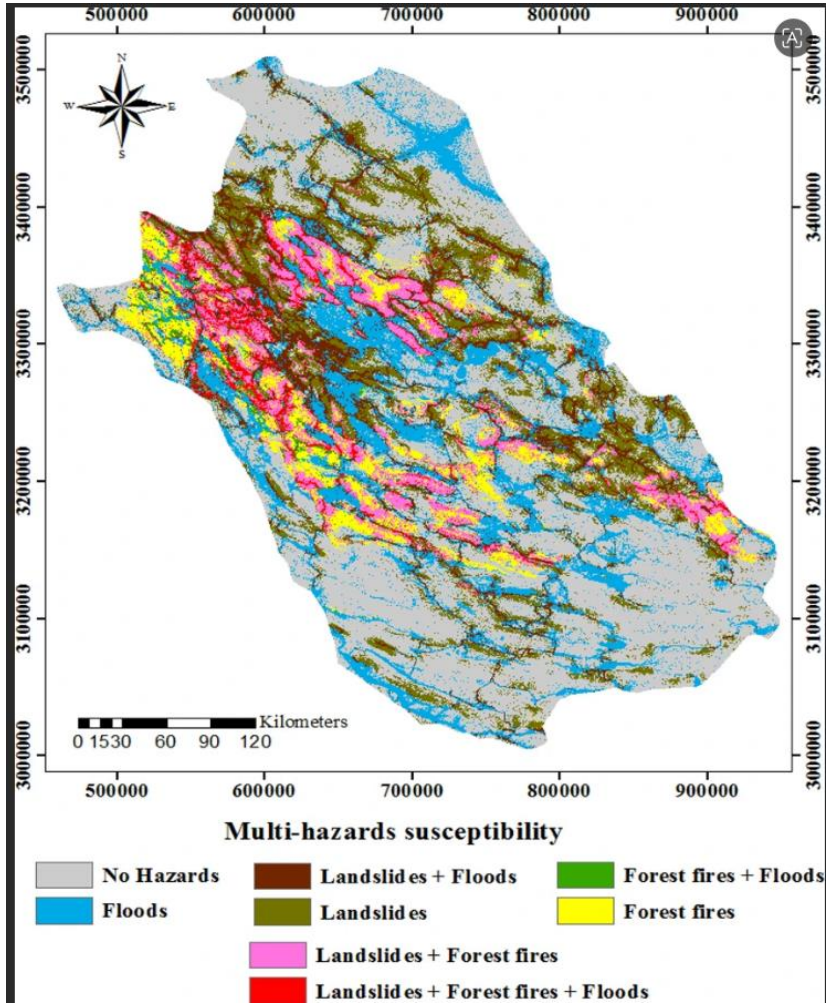
Prototype deliverables and performance metrics

Deliverables

- Working Multi Hazard-VLM prototype with three decoupled task heads (language, coordinates, radius)
- Automated 30-day rolling forecast pipeline with 12–24 hr ingestion cadence
- Interactive H3 heatmap dashboard visualizing daily hazard-risk evolution.
- Expected outcome image from[12]

Performance Metrics

- Disaster caption is measured from damage assessment precision (DAP), damage detail recall (DDR), and factual correctness (FC).
- Restoration advice is measured from recovery necessity (RN), strategic completeness (SC), and action priority precision (APP).
- The average accuracy (AVG) denotes the overall performance. [9]



10. Industry Relevance

Department / Agency	Spatial Output Limitations
NDMA (National Disaster Management Authority)	Does not generate core forecasts; acts as a downstream aggregator of text warnings.
ISRO (Indian Space Research Organisation)	Primarily reactive (maps post-disaster tracks/burn scars) or relies on manual visual analysis.
CWC (Central Water Commission)	Point-Based Spatial Blindness: Predicts water level heights only at specific stationary physical river gauges.
INCOIS (Indian National Centre for Ocean Information Services)	Confined strictly to coastal margins; lacks a framework to track cascading inland hydrological effects.
GSI (Geological Survey of India)	Rely on broad, static regional susceptibility zones; cannot update risk values dynamically day-by-day.

11. Project Timeline

SEM 7 — Implementation Phase

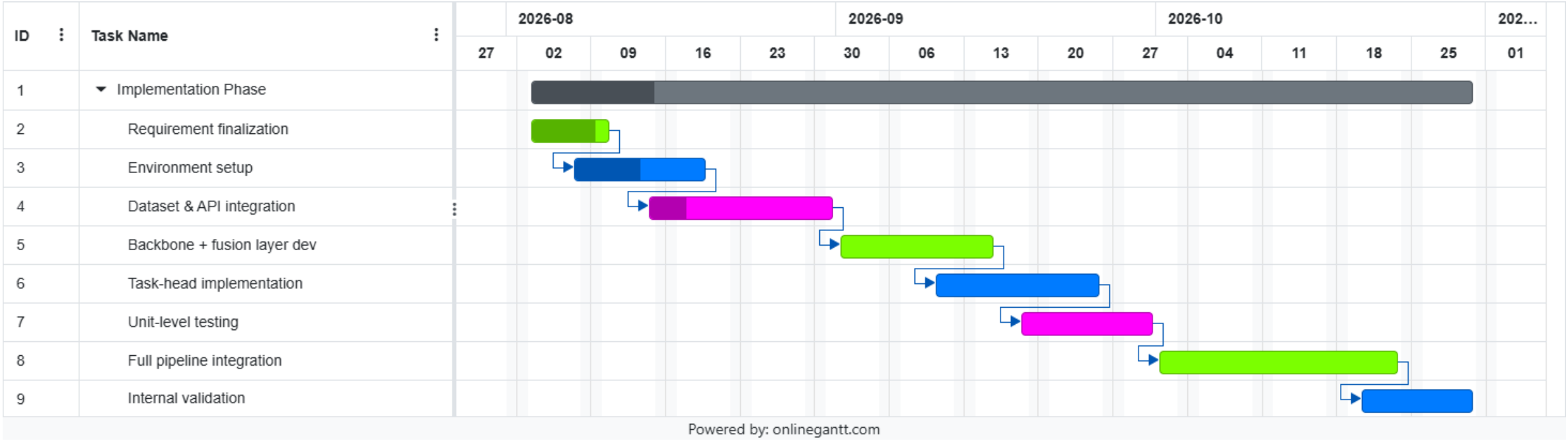
August	September	October
• Requirement finalization • Dataset & API integration • Environment setup	• Backbone + fusion layer dev. • Task-head implementation • Unit-level testing	• Full pipeline integration • Internal validation

SEM 8 — Documentation, Conference & Final Review

Documentation	Conference / Journal	Final Review
• Thesis / project report • Technical documentation • User manual	• Paper drafting • Submission to conference/journal • Reviewer response	• Feedback incorporation • Refinement & optimization • Final demo

11. Project Timeline

Gantt chart of SEM 7



12. Team Roles

Member	Role	Responsibilities
Thanuj	Data & Geospatial	Satellite/radar/weather API integration, dataset curation, H3 grid encoding, data pipeline automation
Krishna	Model Architect (Vision)	Spatiotemporal Mamba backbone, cross-modality alignment, cross-attention fusion layer
Siva Pranav	Model Architect (Language & Regression)	Language reasoner head, spatial coordinate regressor, radius estimator MLP, prompt encoding
Tejadeep	Systems & Deployment	Docker/Kubernetes deployment, edge failover (TensorRT/Jetson), fallback matrix, dashboard & evaluation benchmarking

13. References

[1] "DisasterM3: A Remote Sensing Vision-Language Dataset for Multimodal Disaster Management," Proc. NeurIPS, 2025.

[2] "FloodSense-VLM: A Lightweight Vision-Language Model for Real-Time Disaster Management," Preprint, 2026.

[3] "SeaMo: A Season-Aware Multimodal Foundation Model for Remote Sensing," Preprint, 2025.

[4] "Modality-Resilient Multimodal Earth-Observation Foundation Models under Missing and Corrupted Sensors," Preprint, 2026.

[5] "Mamba Multimodal Disaster Recognition Network (Mamba-MDRNet)," Preprint, 2025.

[6] "Multi-Modal Attention for Automated Disaster Damage Classification," Preprint, 2026.

[7] "A Foundation Model Approach for Disaster Detection from Heterogeneous Data," Preprint, 2026.

[8] "Natural Disaster Detection Using Explainable Deep Learning," Preprint, 2024.

[9] M. Alruqimi, A. Riche, P. Confuorto, M. Guermoui, S. Bianchini, and F. Melgani, "From Landslide Detection to Multi-Source LLM-Based Reporting: A Complete Framework for Rapid Assessment of Post-Disaster Scenarios," *Remote Sensing*, vol. 18, no. 11, art. no. 1821, Jun. 2026, doi: 10.3390/rs18111821.

[10] Maxar Technologies. Open data program, 2025. https://www.maxar.com/open-data.

[11] Jane Palmer. The new science of volcanoes harnesses ai, satellites and gas sensors to forecast eruptions. Nature, 581(7808):256–260, 2020.

[12] H. R. Pourghasemi, N. Kariminejad, M. Amiri, M. Edalat, M. Zarafshar, T. Blaschke, and A. Cerdà, "Assessing and mapping multi-hazard risk susceptibility using a machine learning technique," *Scientific Reports*, vol. 10, no. 1, art. no. 3203, Feb. 2020, doi: 10.1038/s41598-020-60191-3

[13] Zhiliang Peng, Wenhui Wang, Li Dong, Yaru Hao, Shaohan Huang, Shuming Ma, and Furu Wei. Kosmos-2: Grounding multimodal large language models to the world. arXivpreprint arXiv:2306.14824, 2023.

[14] "Multi-Hazard Mapping using Earth Observation data and Multi-Task Learning," *Zenodo*, Apr. 2026, doi: 10.5281/zenodo.19879172.

13. References

[15] M. Wieland, F. Fichtner, S. Martinis, S. Groth, C. Krullikowski, S. Plank, and M. Motagh, "S1S2-Water: A global dataset for semantic segmentation of water bodies from Sentinel-1 and Sentinel-2 satellite images," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 17, pp. 2002–2012, 2024, doi: 10.1109/JSTARS.2023.3333969.

[16] Hanoona Rasheed, Muhammad Maaz, Sahal Shaji, Abdelrahman Shaker, Salman Khan, Hisham Cholakkal, Rao M Anwer, Eric Xing, Ming-Hsuan Yang, and Fahad S Khan. Glamm: Pixel grounding large multimodal model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 13009–13018, 2024.

[17] Zhongwei Ren, Zhicheng Huang, Yunchao Wei, Yao Zhao, Dongmei Fu, Jiashi Feng, and Xiaojie Jin. Pixellm: Pixel reasoning with large multimodal model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 26374–26383, 2024.

[18] Akashah Shabbir, Mohammed Zumri, Mohammed Bennamoun, Fahad S. Khan, and Salman Khan. Geopixel : Pixel grounding large multimodal model in remote sensing. ArXiv, 2025.

[19] Dan H Shugar, Mylène Jacquemart, David Shean, Shashank Bhushan, Kavita Upadhyay, Ashim Sattar, Wolfgang Schwanghart, S McBride, M Van Wyk De Vries, M Mergili, et al. A massive rock and ice avalanche caused the 2021 disaster at chamoli, indian himalaya. Science, 373(6552):300–306, 2021.

[20] Sagar Soni, Akshay Dudhane, Hiyam Debary, Mustansar Fiaz, Muhammad Akhtar Munir, Muhammad Sohail Danish, Paolo Fraccaro, Campbell D Watson, Levente J Klein, Fahad Shahbaz Khan, et al. Earthdial: Turning multi-sensory earth observations to interactive dialogues. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025.

[21] Satellite Based Observations of Forest Fires, 2024 in Uttarakhand - Sudoor Manthan- IIRS Science Portal Sudoor Manthan- IIRS Science Portal